

(a) hybrid artifact

CMS SignedData, detached, one SignerInfo

signedAttrs

contentType
messageDigest = X
bindingAttr = B
(bound modes only)

↓ DER

ECDSA P-256 signature

unsignedAttrs

OID 1.3.6.1.4.1.59999.1
(unregistered)
pk_{pqc}, PQC certificate
mode label
PQC signature over Y
no signature covers this:
deletable, replaceable,
rewritable

the modes differ only here

mode	X: classical digest	Y: PQC signs
concat	H(M)	H(M)
bound-attr	H(M)	B
bound-digest	B	B

B = H(ctx || mode || pk_{classical} || pk_{pqc} || H(M)),
each field length-prefixed

(b) evaluation design

3 PQC algorithms × 3 modes, every cell recorded

unmodified third party

	hybrid	hardened	classical	openssl
intact	•	--	•	RQ1
PQC attr. deleted	•	--	•	attack
PQC half replaced	separates	•	•	•
mode rewritten	downgrade	remedy	•	•
document altered	reject	reject	reject	reject

hybrid: our verifier, full rules. hardened: + mode-consistency check.
classical: RFC 5652 only. openssl: CLI with no OQS provider loaded.

• recorded; boxed: the cell a claim rests on; -- not applicable. Shaded row is a control: if any verifier accepts it, the run aborts and no row is reported.